

The sequence below represents a mature mRNA molecule in a eukaryotic cell:

5'-GCGGACAUGCGGCCACGAUAGACUCCAAAAAAA-3'

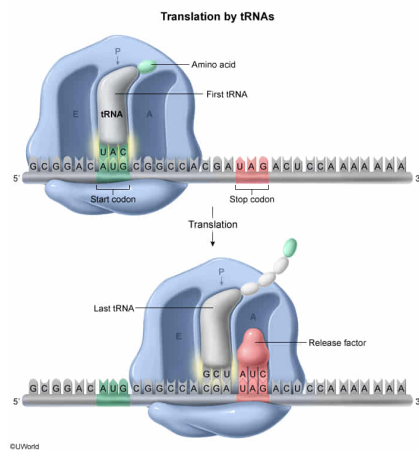
If translation of this mRNA molecule were to occur, which of the following most likely corresponds to the anticodons of the first and last tRNAs that would be brought into the ribosome, respectively?

- ☐ A. 3'-TAC-5' and 3'-GCT-5' [7%]
☐ B. 3'-UAC-5' and 3'-UUU-5' [24%]
☐ C. 3'-CGC-5' and 3'-TTT-5' [25%]
✓ ☒ D. 3'-UAC-5' and 3'-GCU-5' [42%]

Correct

42% answered correctly

Explanation:



Codons are consecutive sequences of three nucleotides in DNA or RNA that together code for either an amino acid or a stop signal that halts translation. Together, these codons comprise the **genetic code**. In **eukaryotic cells**, RNA polymerase in the nucleus **transcribes** DNA genes into pre-mRNA. Pre-mRNA is **processed** into mature mRNA, which is transported out of the nucleus where it can then be translated by the ribosome.

Translation of mRNA occurs in the 5' to 3' direction and is facilitated by **tRNAs**. Each tRNA is "charged" with a specific amino acid and contains an **anticodon** that **base pairs** with a **specific mRNA codon**. Translation initiates at the mRNA **start codon** (5'-AUG-3'), which codes for the amino acid methionine. Translation continues along the **reading frame** until a **stop codon** (UAG, UAA, or UGA) is reached. Stop codons **do not code for amino acids** and are recognized by **release factors** instead of tRNA. Release factors induce the dissociation of the newly translated peptide from the ribosome.

The question presents a mature mRNA molecule that is translated by the ribosome. Because translation of this mRNA **initiates at the 5'-AUG-3'** start codon, the **first tRNA** brought into the ribosome must have the anticodon **3'-UAC-5'**. Translation halts at the stop codon (**5'-UAG-3'**). However, stop codons are recognized by *release factors*, *not* tRNA

molecules. Therefore, the **last tRNA** brought into the ribosome binds the codon **directly upstream** of the stop codon. The final codon *before* the stop codon is **5'-CGA-3'**, so the anticodon of the **final tRNA** must be **3'-GCU-5'**.

(Choices A and C) Thymine (T) is *replaced* by uracil (U) in RNA molecules. Therefore, tRNAs used during translation would *not* contain T nucleotides.

(Choice B) Translation of all mRNA molecules begins at the start codon 5'-AUG-3', which is recognized by the methionine-charged tRNA with the anticodon 3'-UAC-5'. However, translation would be halted at the stop codon (5'-UAG-3'), located *upstream* of the poly-A tail, and which is recognized by a release factor. Accordingly, the last tRNA would bind the codon 5'-CGA-3', which is directly upstream of the stop codon, using the anticodon 3'-GCU-5' (*not* 3'-UUU-5').

Educational objective:

Translation of mRNA by the ribosome initiates at the start codon (AUG), which codes for the amino acid methionine. Translation is terminated at the stop codon, which is bound by a release factor that induces dissociation of the translation complex.

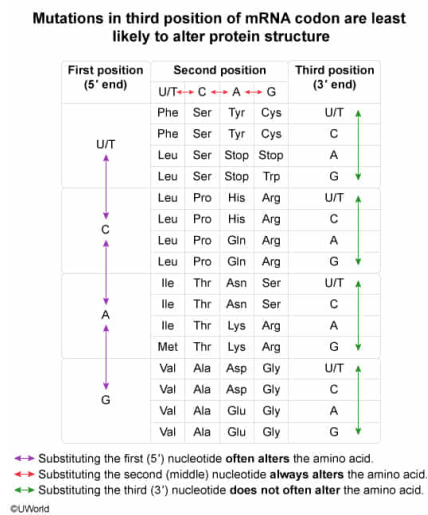
Protein coding genes are composed of multiple codons, sequences of three nucleotides that each code for a specific amino acid. Given that multiple codons may encode a single amino acid, which of the following nucleotide substitutions in a codon would be LEAST likely to affect protein structure?

- ☐ A. Substitution of the nucleotide in the first position of a codon [2%]
- ☐ B. Substitution of the nucleotide in the second position of a codon [3%]
- ☒ C. Substitution of the nucleotide in the third position of a codon [80%]
- ☐ D. Substitution of any one nucleotide within a codon has an equal probability of affecting protein structure. [13%]

Correct

80% answered correctly

Explanation:



Protein-coding genes consist of multiple codons, sequences of three nucleotides that each code for a specific amino acid. Together, 64 total codons comprise the **genetic code**. However, because these 64 codons code for only 20 amino acids, the genetic code is referred to as *degenerate*, meaning **multiple codons** may code for the **same amino acid**.

During **translation**, each mRNA codon (except the stop codon) base-pairs with a corresponding tRNA that is charged with a specific **amino acid**. **DNA mutations** in the protein-coding regions of a gene may alter the sequences of **transcribed** mRNAs, leading to changes in the amino acid structure of encoded proteins. Nucleotides in the first or second position of a codon require traditional **Watson-Crick base pairing** with the tRNA anticodon. Because mutating either of these positions also changes the base pairing, these mutations are likely to alter tRNA binding, and therefore are *most* likely to change the encoded amino acid.

However, **silent mutations** alter the sequences of DNA and mRNA but do *not* affect protein structure. **Mutations** in the **third position** often code for the **same** amino acid, making them silent mutations. This is largely due to less stringent base pairing (**wobble**) at the third position of the mRNA codon and tRNA anticodon. Because mutating nucleotides in the third codon position often result in silent mutations, **substitution** of the **third nucleotide** in a codon is **least likely to affect protein structure**.

(Choices A, B, and D) Because the first two nucleotides in a codon require traditional (Watson-Crick) base pairing with a tRNA anticodon, mutations altering the first or second nucleotide in a codon are *more* likely to alter the amino acid structure of the encoded protein.

Educational objective:

Because multiple codons may code for the same amino acid, the genetic code is called degenerate. This degeneracy is largely due to less stringent (wobble) base pairing between the third nucleotides in the mRNA codon and tRNA anticodon.

Time spent:0 sec
 Subject:
 Biology

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